# Distance sensor

DETECT | PICO BLOCKS | DISTANCE

## What does it do?

An HC-SR04P distance sensor measures how far away something is without touching it.

It works like a bat. One side sends out a click too high for you to hear, the other side listens for the echo, and the time it takes gives the distance.

Your projects can use it to react as something comes closer, or to measure a gap that keeps changing.

## Uses



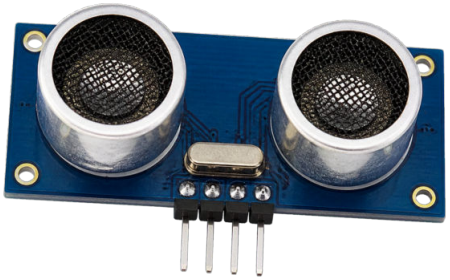
### Parking sensor

DETECT: Gap to the wall → DECIDE: Is it closer than 30 cm? → DO: Beep faster



### Air harp

DETECT: How far your hand is → DECIDE: Which note is that? → DO: Play the note



| VCC | Trig | Echo | GND |
|---|---|---|---|
| Power, 3V3 | Sends a click | Times the return | Ground |



Raspberry Pi Foundation

# Blocks

## MicroLab

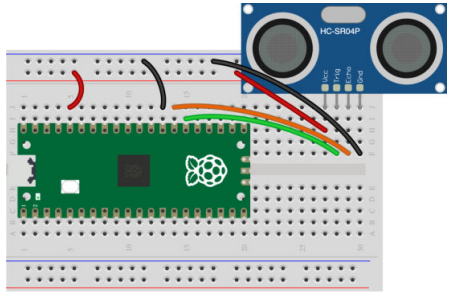
getmicrolab.com




Shows centimetres. 400 means nothing is in range.

# Wiring

**PICO**




HC-SR04P · Trig **GP20** (pin 26) · Echo **GP21** (pin 27) · **3V3 OUT** (pin 36) to top + rail · **GND** (pin 28) to top − rail